PRE-SUBMISSION REPORT

Financial Markets — From Fundamentals to AI-Driven Analytics

1. Project Details

Project Title:

The Role of Geopolitics affecting the Stock market in times of Crisis, especially during War: A Multi-Event Behavioral & Quantitative Analysis (1980–2024)

Project Type:

Research (Quantitative Event Study & Statistical Hypothesis Testing) to ensure both theoretical understanding of the market volatility and sentiment and practical utility for portfolio and risk management during geopolitical conflicts.

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2. Abstract

The aim of this project is to understand how the Indian Stock Market performs in context to the geopolitical tension and crisis, especially during war.

This research investigates whether Indian stock market reaction curves to geopolitical conflicts are determined by war duration, or are primarily driven by economic disruption intensity and investor behavioral responses. Analyzing 21 conflict events (14 global, 7 India-specific) spanning 1980–2024, we employ Event Study Methodology to compute Cumulative Abnormal Returns (CAR) on the NIFTY 50 and BSE SENSEX indices. We examine both direct transmission channels (FII capital Outflows, Brent Crude Oil, Gold) and indirect channels (domestic SIP-driven resilience).

Five formal hypotheses are tested using Welch's t-tests and paired t-tests at the 5% significance level. Key findings: conflict duration does not determine market impact ($p=0.670$); investor overreaction in the first 5 days is 9.7× more intense than sustained response ($p=0.003$); and markets fully adapt within 30 trading days regardless of conflict continuation ($p=0.751$).

3. Problem Statement

Existing literature on war-market relationships often focuses on single events, producing conclusions that are not generalizable.

This study addresses the gap by asking: Do market reaction curves depend on conflict duration (Duration Hypothesis), or are they shaped by disruption intensity and investor psychology (Disruption-Behavioral Hypothesis)?

We test this across 21 diverse conflicts covering 44 years, examining both direct effects (oil shocks, gold flight-to-safety, FII exists) and indirect effects (domestic SIP cushioning).

4. Data Source (27+ years)

All data sourced programmatically via the Yahoo Finance API (yfinance).

Event timeline: 44 years (1980–2024). Continuous price data: 27 years (1997–2024).

Dataset	Ticker	Period	Purpose
NIFTY 50 / SENSEX	^NSEI / ^BSESN	1997–2024	Market index (auto-fallback)
Brent Crude Oil	BZ=F	2007–2024	Direct: Oil shock channel
Gold (COMEX)	GC=F	2000–2024	Direct: Safe-haven flight
USD/INR	INR=X	2003–2024	Currency shock channel
India VIX	^INDIAVIX	2008–2024	Fear/volatility gauge
NIFTY Sectoral Indices	Various	2011–2024	Indirect: Sector rotation

5. Methodology / Plan of Work

Automated 9-script Python pipeline:

- **Data Pipeline:** Download → Clean → Compute daily returns ($R_t = \Delta P_t / P_{t-1} \times 100$).
- **Event Study:** 90-day estimation window → Expected Return → Abnormal Return → CAR at Day +5, +30, +90, +180.
- **Volatility:** σ_war / σ_normal ratio. Threshold: >1.5 = moderate spike, >2.0 = severe.
- **Sectoral:** Cross-sectional heatmap (Bank, Pharma, Energy, FMCG) per conflict.
- **Classification:** A/B/C grading — A: CAR < −5%, B: −5% to −2%, C: > −2%.
- **Hypothesis Testing:** 5 hypotheses via Welch's t-test and paired t-tests (α = 0.05).
- **Behavioral:** Results interpreted through Prospect Theory, Herd Behavior, and Adaptive Market Hypothesis.

6. Progress So Far

- ✓ Data collection & cleaning — all 7 datasets (NIFTY, SENSEX, Oil, Gold, FX, VIX, Sectoral).
- ✓ Percentage impact analysis — 21 events × 8 time windows.
- ✓ Event study (CAR) — 18/21 events analyzed (3 pre-1997 excluded due to data limits).
- ✓ Volatility analysis — σ_war/σ_normal ratios for all 18 events.
- ✓ Hypothesis testing — 3/5 hypotheses supported (H1, H4, H5) at α = 0.05.
- ✓ 8 publication-quality graphs generated at 300 DPI.

7. Planned Next Steps

- Draft Final Report (6 sections: Intro, Literature Review, Methodology, Results, Discussion, Conclusion).
- Create Presentation PPT (15 slides with key graphs and statistical findings).
- Finalize GitHub Repository with README and reproducibility instructions.

8. Expected Outcome / Preliminary Insights

Using time-series analysis of daily returns and variance during defined event windows, the findings indicate that global conflicts provoke prolonged, U-shaped market corrections driven by capital flight, supply-chain constraints, and crude oil price shocks. In contrast, local border conflicts result in minor, short-lived volatility followed by rapid, V-shaped recoveries. Furthermore, the study highlights how structural shifts in domestic retail participation (SIPs) and systematic investments have acted as a buffer, mitigating the impact of international geopolitical tensions on the Indian equity market.

Preliminary statistical results confirm this framework: the initial 5-day shock intensity is 9.7× greater than the sustained 30-day impact (p=0.003), proving investor overreaction (Prospect Theory). Markets stabilize within 30

days regardless of conflict continuation (p=0.751), supporting the Adaptive Market Hypothesis. War duration has no significant effect on market reaction magnitude (p=0.670).

9. References

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[7] Yahoo Finance API — https://finance.yahoo.com/ | yfinance — https://github.com/ranaroussi/yfinance

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